

BEC303 Network Analysis — Assignment 4 — s-domain analysis, transfer functions and synthesis

Module 4: Laplace Transformation and Applications

Field	Value
Maximum marks	20
Weightage	10% of CIE
Issue date	<<TBD>>
Due date	<<TBD>>
Items	10 problems + 3 theory questions
CO / Bloom coverage	CO4: 13 · L1–L2 0, L3–L4 5, L5–L6 8

Objective

Establish the s-domain as the default tool for transient analysis, build fluency in reading circuit behaviour off a pole-zero plot, and introduce synthesis as the reverse of analysis subject to a realisability test.

Learning outcomes

- Student will be able to transform a network including initial conditions and solve it algebraically.
- Student will be able to obtain transfer functions and interpret pole and zero locations physically.
- Student will be able to relate pole position to bandwidth, settling time and damping.
- Student will be able to synthesise a simple network from a specified transfer function and verify it.

Part A — Problems (17 marks)

P1. Transform a second-order differential equation with two non-zero initial conditions, solve algebraically for the output transform, and invert it. Show explicitly where each initial condition entered. [C04 | L4 | 2 marks]

P2. For a circuit with one energised inductor and one energised capacitor, draw the complete s-domain equivalent including both initial-condition sources, and state the polarity justification for each. [C04 | L4 | 2 marks]

P3. Solve the P2 network by s-domain nodal analysis, then separate the partial fraction expansion into forced, natural and initial-condition contributions and label each. [C04 | L5 | 2 marks]

P4. Perform partial fraction expansions of one function with distinct real poles, one with a repeated pole, and one with a complex conjugate pair, and invert all three. [C04 | L3 | 2 marks]

P5. Verify all three of your P4 results using the initial-value theorem, and check the final-value theorem where it is valid. For the case where it is not valid, explain why and show what misleading number it would have returned. [C04 | L5 | 2 marks]

P6. Obtain the transfer function of a second-order passive network, plot its poles and zeros, and predict the step response character before computing it. Then compute it and compare. [C04 | L5 | 2 marks]

- P7.** Synthesise the transform of a trapezoidal pulse of your own specification using shifted steps and ramps, and verify by reconstructing the waveform at five test times. [C04 | L4 | 2 marks]
- P8.** For an RC low-pass of your choice, tabulate pole location, time constant, cutoff in rad/s and hertz, 10–90 % rise time, and the bandwidth–rise time product. Repeat for a second RC with the pole ten times further out. [C04 | L5 | 1 marks]
- P9.** Compute the impulse, step and ramp responses of one circuit and demonstrate the integration relationship between them both graphically and algebraically. [C04 | L4 | 1 marks]
- P10.** Realise a specified first-order transfer function with passive components, then analyse a two-stage unbuffered cascade of it and report how far the realised poles differ from the intended $H(s)^2$. [C04 | L5 | 1 marks]

Part B — Theory (3 marks)

- T1.** Explain why the Laplace lower limit is taken at 0^- rather than 0^+ , and give one concrete case in which the choice changes the answer. [C04 | L5 | 1 marks]
- T2.** Argue why the pole nearest the $j\omega$ axis dominates the response, and state the circumstances under which a higher-order circuit may be safely approximated by its dominant pole alone. [C04 | L5 | 1 marks]
- T3.** Explain why not every rational function can be realised by a passive network, and state the realisability conditions for a driving-point impedance. [C04 | L5 | 1 marks]

Submission guidelines

- Format: PDF, single file
- File name: <USN>_BEC303_Assignment4.pdf
- Channel: LMS upload
- Late policy: 10% per day, nothing after 5 days
- Every numerical answer must carry units and a stated reference direction.

Assessment rubric

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Transform and inverse-transform technique	6	All transforms correct including initial-condition terms; partial fractions handled correctly for distinct, repeated and complex poles; improper fractions divided out first.	Technique sound with one minor residue error.	Distinct-pole cases correct but repeated or complex poles mishandled.	Initial-condition terms dropped; improper fractions expanded without division.

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
s-domain network modelling	5	Every element and initial condition correctly modelled with justified polarity; solution decomposed into forced, natural and initial-condition parts.	Models correct with polarity justified implicitly.	Element impedances correct but initial-condition sources omitted or misplaced.	Capacitor impedance written as sC or initial conditions folded into Z_{th} .
Pole-zero interpretation	5	Response character predicted from pole positions before computation and confirmed afterwards; dominant pole correctly identified; bandwidth and settling time related quantitatively.	Correct interpretation with limited quantitative linkage.	Poles located but their physical meaning stated only loosely.	Largest-magnitude pole assumed dominant; rad/s and hertz confused.
Synthesis and verification	4	Realisability checked, components derived, circuit verified by forward analysis, and loading effects quantified.	Realisation correct and verified; loading discussed qualitatively.	Realisation correct but unverified.	Cascade assumed to multiply without buffering; no realisability check.
Total	20				

Course outcome attainment

Course outcome	Marks carried
CO4	20
Total	20

Academic integrity

Individual submission. Identical submissions score zero for both parties.